Perceptions of a Robot That Interleaves Tasks for Multiple Users

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When robots have multiple tasks to perform, they must determine the order in which to complete them. Interleaving tasks is efficient for the robot trying to finish its to-do list, but it may be less satisfying for a human whose request was delayed in favor of schedule efficiency. Following online research that examined delays with various motivations, we created two in-person studies in which participants' tasks were impacted by the robot's other tasks. In the first, participants either requested a task for the robot to complete on their behalf or watched the robot performing tasks for other people. We measured how their opinions changed depending on whether their task's completion was delayed due to another participant's task or they were observing without a task of their own. In the second, participants had a robot walk them to an office and became delayed as the robot detoured to another location. We measured how opinions of the robot changed depending on who requested the detour task and the length of the detour. Overall, participants positively viewed task interleaving as long as the delay and inconvenience imposed by someone else's task were small and the task was well-justified. Also, observers often had lower opinions of the robot than participants who requested tasks, highlighting a concern for online research.

CCS Concepts: • **Human-centered computing → Empirical studies in HCI**; • **Applied computing → Psychology**;

Additional Key Words and Phrases: Robot task scheduling, off-task behaviors, perceptions of robots

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1 Introduction

As robots become more reliable and affordable, they are increasingly being used to deliver items to people in a wide variety of settings. For example, a number of delivery robots navigate neighborhoods and deliver food and other items to people's homes after orders are placed (e.g., Coco, Daxbot, Kiwibot, Starship).[1] Other delivery robots move around indoor areas and carry items to very specific locations, such as bringing meals to people's tables in restaurants (e.g., temi GO, Richtech Matradee)[2] and transporting medications and supplies in hospitals (e.g., TUG).[3]

As the usage of these robots increases, so too does the need for a method that enables them to decide how to best complete multiple users' tasks. One approach is for robots to execute one task to completion before beginning the next, regardless of how many and what kind of steps or subtasks comprise each task. This sequential method is simple but is often less efficient than *interleaving* the subtasks that comprise the tasks, i.e., performing one subtask at a time for multiple tasks. Interleaving can result in a shorter robot path and/or shorter completion time for the entire task queue. To this end, algorithms for planning to satisfy multiple interacting goals with varying priorities have long been used to create robot task queues (e.g., [9]). Recently, task interleaving has been used in contexts as diverse as multi-robot aircraft manufacturing teams [8, 24], human–robot space teams [22], multi-robot delivery and maintenance teams [19], and multiple robots providing entertainment and reminders in a care home [2].

One important consideration is how humans perceive robots that interleave tasks. For example, Walker et al. [27] uncovered that participants (who were all passive observers) viewed a robot negatively when it interleaved an additional task that was motivated by its curiosity into its original task queue generated by other people. Carter et al. [4] showed that observers who watched a robot perform tasks for multiple other users in sequence found it to be more competent than a robot that interleaved subtasks to complete the tasks. Despite this perception, however, most observers still recommended that the robot interleave subtasks to improve efficiency. That study also found that the observers' satisfaction was impacted by the order in which the interleaving occurred and by how far out of the way the robot went to complete a task that was requested second. Together, these findings suggest that there are contradictions and nuances in how people want robots to queue multiple tasks.

Both of these prior works involved participants watching videos of a robot doing tasks for an on-screen human, not for the participants. The impressions of task owners, who make requests, might diverge from observers, who do not. To address this limitation, we explored how perceptions might change if the participants themselves interact with and task the robot. We performed two experiments that conceptually replicated and expanded upon the previous work but took place in person. This change allowed us to manipulate participant roles and the temporal impacts of task and subtask order. In the first experiment, sessions were designed such that one participant requested a delivery task first, and then a different participant requested an information-gathering task. A third participant observed the entire scene in a role analogous to the online participants in the prior work. We tested several subtask interleaving conditions that varied the task completion order and the amount of waiting time for the participants. Then, we were able to compare how roles and queuing patterns affected impressions of the robot. Next, we wanted to examine tasks with more significant impacts on participants' time and energy. In our second experiment, participants were led to an intended destination ("Stephanie's office") by the robot but were interrupted when it detoured for another task. We manipulated the distance of the detour as well as who requested the

[1] www.cocodelivery.com, daxbot.com, www.kiwibot.com, www.starship.xyz
[2] www.robotemi.com/robots/, www.richtechrobotics.com/matradee
[3] aethon.com/mobile-robots-for-healthcare/

detour task (unspecified, "someone," and "Stephanie"), impacting the explanation that the robot provided for its behavior.

The results of the experiments suggested that short delays for task owners do not have significant negative impacts on perceptions of the robot, but long delays require specific justification to mitigate impacts on perceptions of the robot. Also, observers perceive the robot more negatively than actively involved task owners. Overall, participants overwhelmingly endorse task interleaving in abstract scenarios that do not affect them directly.

2 Related Work

When scheduling for themselves, humans often plan their agendas such that similar tasks are clustered together in time to improve efficiency [13], and switching between different tasks often creates problems (e.g., [6]). To allow robots to create their own schedules, algorithmic methods have sought to learn from human data how to prioritize tasks in queues and determine appropriate time points for robot task switching [7, 20]. However, watching a robot perform multiple tasks or interleave steps of tasks may be confusing for human users and observers. Thus, we are interested in people's perceptions of a robot as it completes multiple tasks from a queue in various orders and degrees of transparency. We also considered each person's relationship to the robot as a user or observer to determine how that perspective impacts impressions.

Sometimes a person anticipates that a robot will complete a specific task and it instead performs an unexpected behavior. In those cases, there are multiple ways to explain the robot's performance: the robot could be completely unmotivated, the robot could be motivated by an unknown internal factor, or the robot could be motivated by a different, externally generated factor (e.g., a task for another user). The human's interpretation of the robot's behavior can impact their impressions of the robot. Early research on internally motivated robot behaviors has looked at how robots could gather additional data on the way to/from performing another task [10, 14, 15], but it did not consider user or observer experience. Recently, Walker et al. [27] used videos to study perceptions of a robot that gathered extraneous data during a requested task, with varied or no explanations provided for the robot's behavior. After seeing these extra behaviors without explanation, human perceptions of the robot's competence decreased. However, the impact was somewhat mitigated if the robot explained that it was curious. Whether the robot went out of its way and potentially inconvenienced the task requester did not affect ratings, despite some negative comments about extended routes.

This work was expanded upon by the introduction of explicitly motivated tasks. Carter et al. [4] performed an online video study that replicated the curiosity and no explanation conditions of Walker et al. [27] and added a new condition: a second request from another person. In that work, participants acknowledged that the robot should interleave tasks for the sake of efficiency, but they rated it as more competent when it performed tasks sequentially, finishing one task before starting the next. Interleaving tasks was also endorsed most strongly when there was a second person making the request for the stop. Participants did not approve of interleaving the second task under other circumstances: when no motivation was given, for the sake of the robot's curiosity, or when it led to long detours from the first task. Overall, external motivation by another user's request impacted perceptions of interleaved tasks. Our first study is an in-person replication of [4] while expanding it to include participants who were not simply observers of the robot's actions. Later, we will discuss the specific details of the changes and the impacts on the results.

One method of illuminating a robot's motivation in ambiguous circumstances is for it to explain its actions to people to convey its goals and plans (e.g., [11, 12, 16]). Early work recommended that robots who needed people to get out of their way should justify the disruption by telling those people their goals and that the people should move [25]. However, it recently was argued that providing task information to other people could violate an initial user's privacy [5]. Robot designers must know what information is important and necessary to convey to people about task performance and interleaving to determine how to best balance these considerations. To this end, we further examined how known external motivation for a robot affects humans' opinions of them. We explored the impacts on impressions of the robot of knowing whether another person requested an additional task and who that person was. We also looked at how the amount of inconvenience introduced by interleaved tasks affected people's perceptions.

3 Experiment 1 Method

This in-person study replicated and expanded upon previous online work [4] aiming to understand how humans perceive a robot that delays completing its original task to interleave others. By performing this study in person, we hoped to increase the participants' engagement in the robot's behavior, have them invest in the task by making requests themselves, and participate in an environment with fewer distractions. Thus, we expected that they were more likely to feel real impact from the robot's task ordering than observers of prerecorded scenes. The experiment involved two requested tasks: *first* and *second*. The first requested task was to bring a box of crackers to the participant requesting them. The second task involved scanning and reporting a box's contents to fulfill a second participant's request. Analyzing people's perceptions of the robot interleaving these tasks for various reasons can then determine the conditions under which interleaving tasks is tolerable.

3.1 Design

This study was a 3 x 4 mixed design with three Role conditions between participants and four Task Order conditions within participants.

The three Role conditions specified the role that each participant had in the interaction. The possible roles were:

- *Requester 1*, referred to as Taylor, who requested that the robot bring them a box of crackers first (the first task);
- *Requester 2*, referred to as Jordan, who made the request that the robot gather information about the contents of a box (the second task); and
- *Observer*, who watched the entire interaction. Each experimental run had between zero and two observers.

All participants saw all four Task Order conditions in an order determined by a balanced Latin Square. The conditions manipulated the inconvenience of the secondary task to the completion of the first task by changing the location of the box to scan. Paths are illustrated in Figure 1.

- *EnRoute*. The robot leaves from its dock near Requester 1 and stops at Box B to scan its contents. Then, it goes to Box E, gets the crackers, and brings them to Requester 1.
- *EnReturn*. The robot leaves from its dock near Requester 1, proceeds to Box E, and grabs the crackers. On its way back to Requester 1, it scans Box B.
- *Detour*. The robot leaves from its dock near Requester 1 and goes to Box G to scan its contents. Then, it goes back to Box E, grabs the crackers, and carries them to Requester 1.
- *Sequential*. The robot leaves from its dock near Requester 1 and goes to Box E to grab the crackers. It brings them back to Requester 1 and then navigates to Box B to scan its contents.

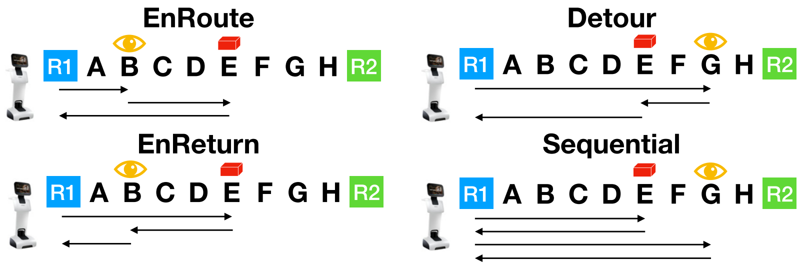

Fig. 1. The four Task Order conditions: EnRoute, EnReturn, Detour, and Sequential. Box E contained crackers. Box B or Box G was surveyed.

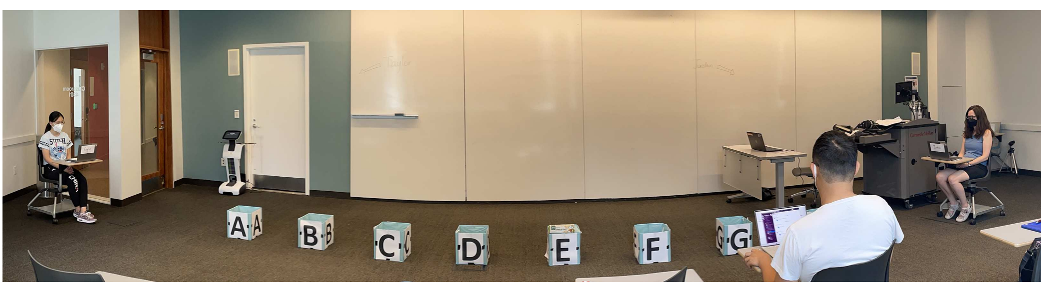

Fig. 2. The setup of the room for Experiment 1, taken using a fisheye lens. Requester 1 sat on the left and Requester 2 on the right. The robot docked near Requester 1. Boxes A to H were on the floor. The observer sat viewing the entire scene (closest to camera).

3.2 Setup

The setup is shown in Figure 2. This research used a temi v2 mobile robot with a supplementary magnetic arm. The robot and its charging station were located in the corner of a university classroom. A desk with a laptop and a chair was placed on one side of the room for the participant who would play the role of Requester 1, and another desk, laptop, and chair were on the other side of the room for the participant who would play the role of Requester 2. Two additional desks, laptops, and chairs were placed in the center rear of the room for participants who would play the role of observers. Two experimenters were seated in the top right corner of the room, where they would not interfere with the robot's movements or the participants' views of the robot and the room. In a line between Requester 1 and Requester 2, we placed a row of eight 12 x 12″ boxes, labeled with letters A to H, on the ground about 18″ apart from each other. Each box contained one or more items. This setup was created to correspond to prior research [4, 27].

Each participant's laptop displayed a task interface that we designed specifically for this experiment (see Figure 3). On the left side of the interface, there were instructions for the participants to follow during the study. On the right side, there were displays specific to each role. The participants role-playing as Requester 1 and Requester 2 were periodically shown a button to click to request a task from the robot. Below the location for that button was a white rectangle that showed a task queue for the robot that displayed the request time, requester, and task for each task that the robot was performing (regardless of who requested it). This interface was developed to maintain performance accuracy, correlate with real-life interfaces (e.g., Starship Deliveries), and align with previous work that did not rely upon accurate natural language interactions with robots (e.g., [4, 17, 26]).

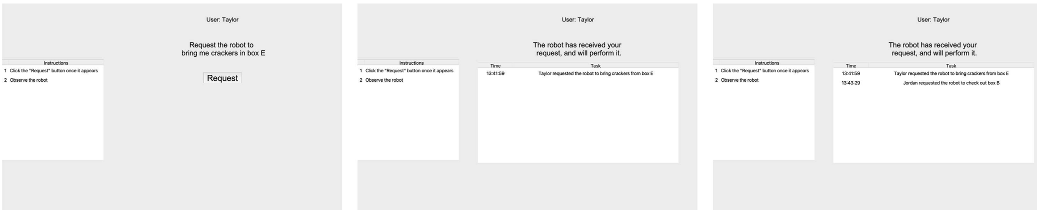

Fig. 3. (Left) When a trial began, Requester 1 (Taylor) saw that they should request their task. (Center) The task request appeared on all participants' interfaces. (Right) Each participant interface also updated with Requester 2's request once made.

3.3 Procedure

Each experiment session contained between two and four participants and two experimenters. When the participants arrived, an experimenter obtained informed consent for the research. Then, participant roles (Requester 1, Requester 2, Observer) were randomly assigned, and people were shown to their appropriate seats. (If only two people arrived for the study, they played the roles of Requester 1 and Requester 2.)

Next, one of the two experimenters introduced the robot and the study. This included a description of the robot's ability to observe the contents of boxes and pick up items from boxes, as well as an explanation of each participant's role. Specifically, the experimenter explained that Requester 1 and Requester 2 were users of the robot and could request that the robots perform tasks for them while the Observers watched. The experimenter also mentioned that all participants should observe both their screens and the robot carefully because they would need to describe what they observed on questionnaires after each trial. Then, the experimenter informed participants that there would be a brief practice trial followed by four trials and that they should remain quiet and raise their hands for any questions.

In the practice trial, Requester 1 and Requester 2 saw instructions to press their request buttons when they appeared on their screens. Requester 1's button appeared on their screen first after the experimenter told them to begin; after pressing it, that participant saw a congratulatory message. Subsequently, Requester 2's button appeared on their screen, and they received an identical message after pressing it. Next, a message appeared on the Observers' screens that both Requester 1 and Requester 2 pressed their request buttons and that the practice trial was finished. The experimenters asked whether there were any questions before reminding them that there would be four trials, each of which would be followed by a questionnaire. The participants were given no specific guidelines on how to evaluate the robot's performance or additional information about the motivation for the task. Notably, they were not explicitly told to want their tasks completed quickly and/or efficiently.

For the subsequent four trials, one experimenter announced the beginning of the trial while the other used a computer to prompt the aforementioned task interface (Section 3.2) to appear for everyone (Figure 3). All participants were shown a task queue with time stamps after Requester 1 and Requester 2 made their requests. Then, an experimenter initiated the robot's performance of the requested actions according to the relevant Task Order condition (EnRoute, EnReturn, Detour, Sequential). When the robot arrived at Box E, it would rotate such that its arm faced the box, and it could use the magnet to pick up a box of crackers with a metal panel inside. At Boxes B and G, the robot would look at the box by angling its screen downward and say, "Scanning." At this time, an experimenter sent a message to Requester 2's computer telling them the contents of the box ("There is candy in the box."). When it was time for the robot to deliver the cracker box to Requester 1, it navigated to them, said, "Here are your crackers," and paused for 10 seconds to allow Requester 1 to retrieve the crackers. At the end of each trial, the robot returned to its docking station, and an experimenter distributed post-trial questionnaires to the participants. After the fourth trial, participants were also given a final questionnaire to complete.

3.4 Measures

Participants completed a questionnaire after each of the four trials as well as a questionnaire at the end of the entire experiment. The questionnaires varied slightly based on whether the participant played the role of Requester 1, Requester 2, or Observer.

3.4.1 Post-Trial Questionnaire. First, participants answered a series of questions related to facts about what had happened in the preceding trial. To check whether they paid attention, they were asked to describe the events of the previous trial and then number the steps that the robot took (either 3 or 4) in the order in which they occurred. Then, the participants who played the roles of Requester 1 and Requester 2 were asked to rate on a 7-point scale (1 = unsatisfied, 7 = satisfied), "Based on the requests you made of the robot, how do you feel about the order in which the robot performed its tasks?" All participants were asked to use the same rating scale to answer, "Based on everything you know about the robot's task list, how do you feel about the order in which the robot performed its tasks?"

We also replicated the rating items from prior work [4, 27], involving 20 single-word items from a combination of the Godspeed questionnaire [1], the robotic social attributes scale [3], and Carter et al.'s [4] novel items. We randomized whether the positive or negative attribute for each item was on the left side. Previously, 18 of the items were found to load onto factors about how competent, observant, and curious the robot was perceived to be; the other two items related to interactivity and intrusiveness [4]. While curiosity was not provided as an explanation for the robot's behavior in this work, we continued to use the curious, inquisitive, and humanlike items from the curiosity factor to align our measures with previous work and with Experiment 2.

3.4.2 Post-Experiment Questionnaire. After completing all four trials, participants filled out a final questionnaire. In it, participants were asked to provide their ideal ordering of steps for the robot for two different potential scenarios. They were given the instructions: "In your opinion, how should the robot complete its tasks when both Taylor and Jordan have a request? (Number from 1 to 4 your preferred order.) Remember that Taylor is next to Box A and Jordan is next to Box H and that the robot starts near box A."

— *Scenario 1*: Taylor requests an item from Box E. Shortly thereafter, Jordan requests to know the contents of Box B.
    (a) Go to Box E
    (b) Alert Jordan about the contents of Box B
    (c) Return to Taylor to give them the item from Box E
    (d) Go to Box B
— *Scenario 2*: Taylor requests an item from Box E. Shortly thereafter, Jordan requests to know the contents of Box G.
    (a) Go to Box E
    (b) Alert Jordan about the contents of Box G
    (c) Return to Taylor to give them the item from Box E
    (d) Go to Box G

Participants were also asked whether a robot should be allowed to complete tasks *not* in the order they are received, and whether a robot should be allowed to interrupt a task to complete another task if it means it can complete all of them faster. This final questionnaire paraphrased items used by Carter et al. [4] to make them appropriate for this experiment.

3.5 Participants

Participants were recruited using a combination of methods: a university-based research recruitment website, posters mounted on campus bulletin boards, and word of mouth. To participate, participants had to be 18 years of age or older and have normal or corrected-to-normal hearing and vision. Ninety people successfully completed the study. This research was approved by our Institutional Review Board, and participants received $8 USD for half an hour of their time.

3.6 Hypotheses

Based on our measures, we hypothesized that:

— (H1a) Requester 2 will be least satisfied with the Sequential path (Measures: "Based on the requests you made of the robot, how do you feel about the order in which the robot performed its tasks?" and "Based on everything you know about the robot's task list, how do you feel about the order in which the robot performed its tasks?")
— (H1b) Requester 1 will be least satisfied with the Detour path. (Measures: Same as H1a)
— (H2) Participants will find the robot taking the interleaved paths to be more efficient and competent. (Measures: Capable, Responsive, Interactive, Reliable, Competent, Knowledgeable, Efficient, Effective, and Focused items as well as *Competent* factor, per [4])
— (H3) Participants will recommend that a robot should be efficient by interleaving tasks rather than sequentially completing tasks. (Measures: Post-experiment questionnaire)

Hypothesis 2 conflicted with previous findings [4]. We expected that in-person participation would result in participants valuing efficiency more because efficiency would have a more direct impact on their time and experience relative to viewers at home computers watching prerecorded stimuli.

4 Experiment 1 Results

Of the 31 sessions, three sessions had only Requester 1 and Requester 2 participants, and the rest had Requester 1, Requester 2, and one or two Observers. A total of five trials were dropped (one from each of five sessions) from the dataset due to robot errors. All participants passed attention checks.

We used the **REstricted Maximum Likelihood (REML)** method to fit a linear model [21, 23] in JMP software [18] to items and included participant number as a random effect. The output of this test is an F-value as the test statistic, a p-value, and power (representing the effect size). Subsequently, we used Tukey's **Honestly Significant Difference (HSD)** tests to look at pairwise comparisons within significant main effects. Significant differences were identified using a threshold of $\alpha < 0.05$. Results with power analyses are reported in Tables 1 and 2. We also performed an exploratory factor analysis with maximum likelihood and an oblique rotation (quartimax) because the ratings may have been correlated. We looked for three factors following previous results [4], and we assigned items to the factor with the maximum correlation, with a minimum requirement for a correlation set at 0.3. After this assignment, one of the three factors did not have any items in it, resulting in two factors with items that related to the robot's Competence and Curiosity. One item (intrusive–unintrusive) did not have any strong correlations with any factor and was not included in related analyses. The results of our factor analysis are available in Table 3. Analyses that used these factors utilized a mean (M) of the associated item scores.

Table 1. For Each Item, the Significance of the Task Order Main Effect, F Ratio, the power of the Main Effect, and the Mean Ratings and Standard Errors for Each Condition

Adjectives	Order p	F	Pwr	EnRou M (SE)	EnRet M (SE)	Det M (SE)	Seq M (SE)
capable	0.3896	1.01		1.70 (0.10)	1.58 (0.10)	1.71 (0.10)	1.83 (0.10)
responsive[a]	0.9069	0.18		4.30 (0.10)	4.31 (0.10)	4.25 (0.10)	4.22 (0.09)
interactive	0.4467	0.89		2.97 (0.27)	2.42 (0.27)	2.69 (0.27)	2.52 (0.12)
reliable	0.7505	0.40		1.69 (0.09)	1.74 (0.10)	1.77 (0.10)	1.83 (0.09)
competent[a]	0.1976	1.57		4.22 (0.10)	4.21 (0.10)	4.20 (0.10)	3.97 (0.09)
knowledgeable	**0.0151**	3.53	0.78	2.09 (0.11)[A]	2.19 (0.11)	2.31 (0.11)	2.57 (0.11)[B]
efficient[a]	**< 0.0001**	30.11	1.00	4.20 (0.11)[A]	4.17 (0.12)[A]	4.07 (0.12)[A]	2.92 (0.11)[B]
effective	0.0687	2.39		1.87 (0.12)	1.86 (0.12)	1.76 (0.12)	2.18 (0.11)
focused	0.3644	1.06		1.76 (0.10)	1.67 (0.10)	1.92 (0.10)	1.78 (0.10)
responsible	0.801	0.33		1.70 (0.10)	1.75 (0.10)	1.74 (0.10)	1.83 (0.10)
intelligent[a]	**0.0017**	5.15	0.92	3.91 (0.12)[A]	3.84 (0.011)[A]	3.67 (0.11)	3.36 (0.11)[B]
inquisitive[a]	0.5291	0.74		3.09 (0.12)	3.06 (0.012)	3.01 (0.12)	2.86 (0.12)
curious	0.2209	1.48		3.09 (0.13)	2.96 (0.13)	3.35 (0.13)	3.14 (0.13)
like	0.0817	2.26		2.00 (0.97)	2.01 (0.10)	2.15 (0.10)	2.31 (0.10)
unintrusive[a]	0.2079	1.52		3.87 (0.12)	3.80 (0.12)	3.55 (0.12)	3.67 (0.12)
humanlike[a]	**0.0066**	4.15	0.85	2.49 (0.12)[A]	2.50 (0.12)[A]	2.44 (0.12)	2.00 (0.12)[B]
careful	0.4195	0.94		1.88 (0.10)	1.96 (0.10)	2.99 (0.10)	2.11 (0.10)
on-task[a]	0.5726	0.67		4.49 (0.09)	4.35 (0.09)	4.31 (0.09)	4.37 (0.09)
attentive	0.4381	0.91		2.02 (0.10)	2.04 (0.10)	2.05 (0.10)	2.22 (0.10)
observant[a]	**0.0453**	4.05	0.66	3.90 (0.11)	3.94 (0.11)	3.82 (0.11)	3.53 (0.11)

Significant main effects are denoted in bold. Significant pairwise comparisons ($p < 0.05$) are denoted by the superscripts A and B, in which a condition noted with an A is significantly different than one noted with a B.
[a]For half of the adjective pairs, the word order was reversed so that the negative term was on the left (1) of the scale: unresponsive, incompetent, inefficient, etc. Det, Detour; EnRou, EnRoute; EnRet, EnReturn; M, mean; Seq, Sequential.

Path Satisfaction. First, we examined the responses from the participants playing the roles of Requester 1 and Requester 2 about their feelings regarding the order in which tasks were completed based on *their own task requests.* There was a significant interaction between Task Order and Role ($F = 6.7792, p = 0.0002$), and pairwise comparisons determined that Requester 2 in the Sequential order condition was significantly more unsatisfied than Requester 1 in the Sequential condition or Requester 2 in any other Task Order. When we explored all participants' responses to their feelings about the robot's Task Order based on *everything they knew about the robot's task list*, there was a main effect of Task Order ($F = 11.2117, p < 0.0001$), such that pairwise comparisons showed that Sequential had lower rates of satisfaction than all other Task Orders. Together, these results suggest that the Sequential Task Order is less preferred overall, but specifically so for the person making the second request.

Rating Items. Next, we analyzed the responses from the 20 single-word rating items. For the Task Order variable, we found significant main effects of order on ratings for *knowledgeable, efficient, intelligent, humanlike,* and *observant.* There was also a significant main effect for the c*ompetent factor* ($F = 4.0457, p = 0.0076, power = 0.852$). Pairwise comparisons showed that Sequential was rated less *knowledgeable* than EnRoute and less *intelligent* and *humanlike* than EnRoute and EnReturn. Additionally, Sequential was lower than EnRoute and EnReturn for the c*ompetent factor* overall. Sequential was also rated as less *efficient* than all other orders. There were no significant pairwise comparisons for *observant.*

Table 2. For Each Item, the Significance of the Role Main Effect, F Ratio, the Power of the Main Effect, and the Mean Ratings and Standard Errors for Each Condition

Adjectives	Role p	F	Power	R1	R2	Observer
capable	**0.0127**	4.43	0.76	1.54 (0.09)^{A}	1.71 (0.09)	1.91 (0.09)^{B}
responsive[a]	**0.0204**	3.94	0.71	4.45 (0.08)^{A}	4.19 (0.08)	4.16 (0.09)^{B}
interactive	**0.0224**	3.84	0.69	2.40 (0.10)^{A}	2.77 (0.10)^{B}	2.44 (0.11)
reliable	**0.0018**	6.42	0.90	1.64 (0.08)^{A}	1.66 (0.08)^{A}	2.01 (0.09)^{B}
competent[a]	0.0958	2.36		4.26 (0.08)	4.16 (0.08)	4.01 (0.08)
knowledgeable	**0.0092**	4.76	0.79	2.08 (0.09)^{A}	2.34 (0.09)	2.49 (0.10)^{B}
efficient[a]	**0.0012**	6.90	0.92	4.08 (0.10)^{A}	3.82 (0.10)	3.55 (0.10)^{B}
effective	0.3048	1.19		1.80 (0.10)	1.97 (0.10)	2.01 (0.11)
focused	**0.0280**	3.62	0.67	1.64 (0.09)^{A}	1.75 (0.09)	1.97 (0.09)^{B}
responsible	**0.0243**	3.76	0.68	1.61 (0.08)^{A}	1.75 (0.08)	1.94 (0.09)^{B}
intelligent[a]	0.0755	2.60		3.84 (0.09)	3.68 (0.09)	3.53 (0.10)
inquisitive[a]	0.4030	0.91		3.00 (0.10)	2.91 (0.10)	3.11 (0.11)
curious	0.6015	0.51		3.05 (0.11)	3.21 (0.11)	3.16 (0.12)
like	0.1395	0.14		2.07 (0.08)	2.05 (0.08)	2.26 (0.09)
unintrusive[a]	< **0.0001**	9.71	0.98	3.56 (0.10)^{B}	4.08 (0.10)^{A}	3.50 (0.11)^{B}
humanlike[a]	0.1230	2.11		2.52 (0.10)	2.26 (0.10)	2.27 (0.11)
careful	**0.0001**	0.42	0.98	1.79 (0.09)^{A}	1.91 (0.09)^{A}	2.30 (0.09)^{B}
on-task[a]	0.2272	1.49		4.49 (0.08)	4.35 (0.08)	4.29 (0.08)
attentive	**0.0028**	5.99	0.88	2.08 (0.08)	1.99 (0.09)^{A}	2.31 (0.09)^{B}
observant[a]	0.2244	1.50		3.92 (0.09)	3.75 (0.09)	3.70 (0.10)

Significant main effects are denoted in bold. Significant pairwise comparisons ($p < 0.05$) are denoted by the superscripts A and B, in which a condition noted with an A is significantly different than one noted with a B.
[a]For half of the adjective pairs, the word order was reversed so that the negative term was on the left (1) of the scale: unresponsive, incompetent, inefficient, etc.

For the Role conditions, there were significant main effects of Role on ratings for *capable, responsive, interactive, reliable, knowledgeable, efficient, focused, responsible, intrusive, careful,* and *attentive*, as well as on the overall c*ompetent factor* ($F = 7.0338$, $p = 0.0010$, $power = 0.928$). Pairwise comparisons revealed that Requester 1 rated the robot more *capable, responsive, knowledgeable, efficient, focused,* and *responsible* than the Observer, as well as higher for the c*ompetent factor*. For *interactive*, Requester 1's rating was significantly higher than Requester 2's. For *reliable* and *careful*, Requester 1 and Requester 2 both had higher ratings than the Observer. Requester 1 and the Observer rated the robot as less *unintrusive* than Requester 2 did. Finally, Requester 2 rated the robot as more *attentive* than the Observer did. Notably, we saw no significant effects of role for the c*urious factor* or the individual items for *inquisitive* or *curious*. This was somewhat unsurprising; unlike previous work [4, 27], we did not manipulate the robot's given motive. No interaction effects between Task Order and Role were found.

Final Questionnaire Reflections. For the final questionnaire, participants began by creating ideal orders for scenarios in which Requester 2 requested to know the contents of a box that was either along the route or out of the way relative to completion of the first request. When Requester 2 hypothetically requested information that was along the original route of the first request (Scenario 1), 43 participants selected the EnRoute option, with 41 recommending telling Requester 2 the information as soon as they arrived at Requester 2's box and 2 telling Requester 2 after delivering

Table 3. Factor Loading to Each of the Three Factors

Item	F1	F2	F3	Result
Capable—Incapable	0.71	0.12	-0.15	COMP
Unresponsive—Responsive	-0.55	0.05	0.00	COMP
Interactive—Not Interactive	0.16	-0.18	0.01	COMP
Reliable—Unreliable	0.77	0.10	-0.03	COMP
Incompetent—Competent	-0.71	0.05	0.32	COMP
Knowledgeable—Unknowled.	0.49	-.40	-0.09	COMP
Inefficient—Efficient	-0.57	0.16	0.31	COMP
Effective—Ineffective	0.63	0.02	-0.10	COMP
Focused—Unfocused	0.62	0.14	0.33	COMP
Responsible—Irresponsible	0.64	-0.07	0.18	COMP
Unintelligent—Intelligent	-0.60	0.48	0.23	COMP
Uninquisitive—Inquisitive	-0.23	0.67	-0.14	CRS
Curious—Incurious	0.13	-0.57	0.06	CRS
Like—Dislike	0.60	-.28	0.01	COMP
Intrusive—Unintrusive	-0.21	-0.18	0.00	N/A
Machine-like—Humanlike	-0.16	0.51	0.14	CRS
Careful—Careless	0.53	-0.18	0.28	COMP
Off-task—On-task	-0.53	-0.25	-0.25	COMP
Attentive—Inattentive	0.56	-.16	0.13	COMP
Unobservant—Observant	-0.57	0.39	0.04	COMP

(F1 COMP, F2 CRS, F3 Unlabeled) and assignment by item (COMP = Competence, CRS = Curiosity, N/A = None).

Requester 1's item (see Table 4). Then, 35 participants endorsed the EnReturn strategy, with 30 telling Requester 2 immediately and 5 telling Requester 2 after Requester 1's delivery. Only six recommended performing the tasks in Sequential order. The remaining six participants suggested idiosyncratic orders. For the scenario in which Requester 2's request was out of the way relative to the route for performing Requester 1's task (Scenario 2), 76 participants recommended pathways similar to our Detour condition: 37 said to go out of the way to do Requester 2's task before picking up Requester 1's item, 35 said to go out of the way after picking up Requester 1's item, and 4 recommended paths where the detour happened but Requester 2 didn't get their information until after Requester 1's item was delivered. Only nine participants recommended the Sequential order, and the remaining five participants generated unique orders.

For the question "Should a robot be able to complete tasks not in the order they are received?," 82 participants said yes and 8 said no. Using a Chi-square test, we found no significant main effect of the order of Task Order conditions or Role. When asked, "Should a robot be allowed to interrupt a task to complete another task if it means it can complete all of them faster?" 74 participants said yes and 16 said no (see Table 5). A Chi-square test revealed a significant main effect of the order in which the Task Order conditions were seen ($\chi^2 = 8.210, N = 90, df = 3, p = 0.0419$, effect size w (computed through Cramer's V) = 0.302 (medium-to-large)), but no significant main effect of Role. For Task Order, a significant pairwise comparison was found between seeing Sequential and Detour last ($\chi^2 = 6.966, N = 47, df = 1, p = 0.0083$, w (computed through ϕ) = 0.385 (medium-to-large)), as well as between seeing Sequential and EnRoute last ($\chi^2 = 4.545, N = 50, df = 1, p = 0.033$, w (computed through ϕ) = 0.301 (medium-to-large)).

Table 4. Percentage of Participants Who Recommended Orders Similar to Each Task Order Condition for Two Hypothetical Scenarios

Recommended Order	Scenario 1 (%)	Scenario 2 (%)
Sequential	6.67	10
EnReturn	38.89	—
EnRoute	47.78	—
Detour	—	84.44
Other	6.67	5.56

In Scenario 1, they were asked how the robot should incorporate a second task for Requester 2 that was along the path for an existing task for Requester 1. In Scenario 2, they were asked how the robot should add Requester 2's task when it was out of the way of Requester 1's task.

Table 5. Percentage of Participants Who Said That the Robot Should Not Interleave Tasks ("No") Based on Which Condition They Saw Last

Last Seen	Proportion "No" (%)
Sequential	4.00
EnReturn	11.11
EnRoute	24.00
Detour	31.82

Participants who saw Sequential last were significantly less likely to say that the robot should not interleave tasks than participants who saw Detour or EnRoute last.

5 Experiment 1 Discussion

Overall, participants had varied feelings about the task orders. We found support for our hypothesis (H1a) *Requester 2 will be least satisfied with the Sequential path*. Participants in the Requester 2 role rated the Sequential path as significantly less satisfying than the other paths. Moreover, Requester 2 was significantly less satisfied with the Sequential path than Requester 1 was. In the Sequential condition, Requester 2's task was greatly delayed by the robot navigating across the room three times to complete Requester 1's task first. Our hypothesis (H1b) that *Requester 1 will be least satisfied with the Detour path* was not confirmed: there were no significant effects found for Requester 1's preferences. This contrast between findings may be because Requester 1's request was only slightly delayed by Requester 2's task.

For (H2) *Participants will find the robot taking the interleaved paths to be more efficient and competent*, we found that participants overwhelmingly rated at least one of the interleaved paths as significantly more knowledgeable, efficient, intelligent, and humanlike than Sequential. Additionally, the EnRoute and EnReturn paths had significantly higher *Competence factor* scores than Sequential. These findings partially support H2.

(H3) stated *All participant roles will recommend that a robot should be efficient by interleaving tasks rather than sequentially completing tasks.* An overwhelming majority of participants recommended paths for two scenarios that interleaved the tasks, supporting H3. Within the recommendations for a detour task, participants were split almost evenly (37 to 35) on whether the robot should go out of the way for Requester 2's task before or after picking up Requester 1's item.

This study replicated previous work [4] while expanding it to include participants who were not simply observers of the robot's actions. Also, this research took place in person, potentially impacting participant engagement. While there were observers, there were also task requesters. These changes to the study did affect some results. For example, the previous paper [4] reported that observers rated the robot taking the Sequential path as more competent and observant than when it took other paths. In contrast, we found that taking the Sequential path was rated by all of the current participants as less knowledgeable, humanlike, and intelligent than EnRoute and less intelligent than EnReturn. Moreover, there were no significant interactions between Task Order and Role. We do not know why the experience of passive observation in person produced different results than online observation. Future work could examine this issue in more depth. In another aspect, both sets of research findings concurred: participants recommended that the robot use efficient paths rather than sequential paths. Together, these findings suggest that being an observer in person does not always produce ratings that align with being an online observer, although there are some similarities. The differences potentially are due to varied levels of participant engagement and interest.

6 Experiment 2 Method

While the first experiment focused heavily on different methods for and perceptions of interleaving tasks, the second experiment more deeply considered how a robot's explanations for its actions affect people's perceptions of it. Once again, we sought to increase the participants' realistic experience of the impacts of the robot's behavior relative to previous work [4, 27] by performing the study in person. Additionally, the participants faced more potential inconvenience from the robot's decisions relative to Experiment 1 or previous work [4, 27] by being required to move around and wait for the robot as it performed its tasks. Finally, the impact of having a second requester was altered because the requester was not in the presence of the participant. The requester's identity was also varied. Here, the robot "helped" participants by leading them to an unfamiliar office. While leading them, the robot stopped to inspect the health of a potted plant, explaining the stop in different ways.

6.1 Design

This study had a between-participants design with six conditions (2 Paths x 3 Motives). Each participant or group was assigned randomly to one condition. The two Path conditions included visiting plants either *Near* to or *Far* from the shortest path to the office. See Figure 4 for illustrations of the paths. The three Motive conditions were, along with the corresponding robot speech:

— *No Motive*: "I'm sorry; it will only take a minute."
— *Someone Asked*: "I'm sorry; it will only take a minute. I need to check a plant for someone."
— *Stephanie Asked*:[4] "I'm sorry; it will only take a minute. Stephanie asked me to check this plant."

Thus, we tested the degree to which having a known motivation for a behavior that was or was not requested by a specific person affected participants' perceptions of the robot.

6.2 Procedure

Participants were recruited in the building where the experiment took place and advised that they would be met by an experimenter upon arriving on the correct floor of the office building via the

[4]Note that participants do not know who Stephanie is.

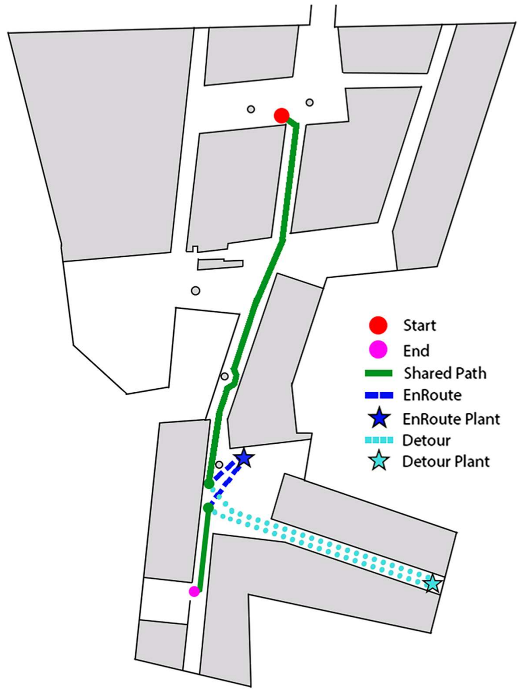

Fig. 4. The robot navigated along the solid green path from the start (red dot) until it either completed the Near detour (dark blue dashes) or Far detour (light blue dots). After the detour, the paths rejoined along the green path to end at the goal location (pink dot).

elevators. Because the floor plan for the building is unusual, this is not an abnormal practice for researchers. The experimenter acquired informed consent.

The experimenter then introduced the temi V2 robot, "We have a robot named Algo that has wheels to navigate from one location to another and a screen at the top. Algo can lead people to places; Algo can also observe the condition of plants. For today's experiment, we're studying how people perceive robots that guide people. All of you will be guided by Algo to Stephanie's office where you will be given a survey." No incentives for quick travel were mentioned. The robot then proceeded to lead the participant(s) toward the office, deviating Near or Far to pause in front of one of the many potted plants distributed along the hallways of the building. While paused, it voiced one of the three explanations (No Motive, Someone Asked, or Stephanie Asked) described above. After 5 seconds, it continued along its path to Stephanie's office. In total, the trip lasted approximately 100 seconds (Near) or 145 seconds (Far).

Upon arriving at their destination, the robot announced, "We have arrived at Stephanie's office." A second person, Stephanie, met them there. Participants then completed a questionnaire about their experience were debriefed by the experimenter and told that the study was finished, and completed a form to receive payment.

6.3 Measures

Participants completed one questionnaire that included four rating items based on prior work [4]:

—"How satisfied were you with the robot's task performance as it brought you to the destination?" (1 = Very Unsatisfied, 5 = Very Satisfied), and "Please rate the robot on the following three items (from 1 to 5)":

— Competent (1) to Incompetent (5),
— Curious (1) to Incurious (5), and
— Observant (1) to Unobservant (5).

They also provided their age and field of study or profession.

6.4 Participants

Participants were recruited as they traveled or were situated in the building in which the study took place. In total, 180 people successfully completed the study. They ranged in age from 18 years to 67 years (M = 29.6, SD = 11.6). The research took place in a building housing **computer science (CS)** offices and classrooms. Ninety-eight participants identified themselves as students in CS or robotics fields, 23 were students studying engineering, 14 people worked in/studied administration or management, 9 were professors or teachers, 8 studied or worked in business fields, and 8 were students with unspecified majors. The remaining 20 participants held a variety of occupations. Overall, the majority of the participants had roles or fields of study that would provide at least some familiarity with artificial intelligence and robots. This research was approved by our Institutional Review Board, and participants were compensated 1 USD for their time.

6.5 Hypotheses

We hypothesized that user satisfaction ratings would reflect that:

— (H4) Participants will be less satisfied with Far detours than Near ones.
— (H5) Participants who hear No Motive for the detour will be less satisfied than those who receive a motive.
— (H6) Participants who hear the motivation that Someone requested a task will be less satisfied than those participants who hear it attributed to Stephanie.

We anticipated that participants who spent significant time and physical effort following the robot to an out-of-the-way place would have more negative views of their experience due to the inconvenience. However, we expected that explicitly stated motives—particularly those arising from a person at the final destination—would moderate negative effects of detours.

7 Experiment 2 Results

As with Experiment 1, we used a REML method to fit a linear model to examine participants' questionnaire results. We then employed Tukey's HSD tests to examine pairwise comparisons within significant main effects. We used a significance threshold of $\alpha < 0.05$ for all tests.

For the rating item on satisfaction with the robot's performance while guiding the participant to the destination, there was a significant main effect of path ($F = 10.4325, p = 0.0015, power = 0.8947$), with the *Near* path rated as more satisfying than the *Far* path. There was also a significant main effect of the interaction between path and explanation ($F = 7.7666, p = 0.0006, power = 0.9477$) (Figure 5). Essentially, Near with any explanation was generally rated as significantly more satisfying than Far with an equivalent or less vague explanation. Specifically, Near with No Explanation was rated as significantly more satisfying than Far with No Explanation; Near with Someone Asked was significantly higher than Far with No Explanation and Far with Someone Asked; and Far with Stephanie Asked was significantly higher than Far with No Explanation and Far with Someone asked.

For ratings of competence, there was a significant main effect of path ($F = 9.8491, p = 0.0020$, $power = 0.8772$), with the *Near* path rated as more competent than the *Far* path. Again, there

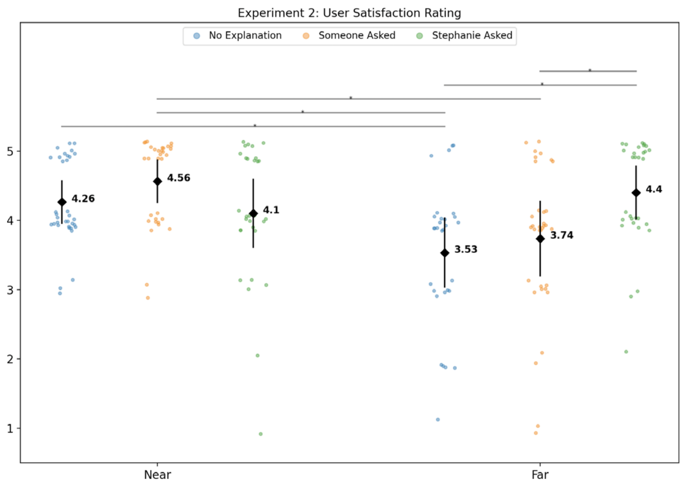

Fig. 5. Satisfaction ratings and standard error for the six conditions in Experiment 2. Statistical significance ($\alpha < 0.05$) between conditions on each measure is denoted by a line connecting the two and an $*$.

was a significant main effect of the interaction between path and explanation ($F = 6.1785, p = 0.0026, power = 0.8871$). Near when Someone Asked received significantly higher ratings than Far with Someone Asked or Far with No Explanation, and Near with No Explanation and Far with Stephanie Asked were rated significantly higher than Far with No Explanation.

For ratings of curiousness, there was a significant main effect of explanation ($F = 5.7320, p = 0.0039, power = 0.8614$) in which No Explanation and Stephanie Asked were rated as significantly less curious than Someone Asked. No other effects were found.

For how observant the robot was perceived to be, there was a significant interaction between path and motive ($F = 6.4764, p = 0.0019, power = 0.9018$) in which Near with Someone Asked was rated as significantly more observant than Far with No Explanation.

8 Experiment 2 Discussion

We assessed our hypothesis (H4): *Participants will be less satisfied with Far detours than Near ones.* We found that participants gave Near detours a significantly higher satisfaction rating than Far detours, supporting H4. Overall, the ratings may be unusually high due to the novelty effect: many visitors to the university who had never significantly interacted with a robot participated. Some people particularly noted the cuteness of the robot as they filled out the survey: "On one hand that was a very inefficient route, on the other [robot name] was cute." Future work is needed to investigate the novelty effect on these ratings and whether there is a larger distinction between the two conditions under other conditions.

For (H5/H6), *Participants who hear No Motive for the detour will be less satisfied than those who receive a Motive* and *Participants who hear a Motive that Someone requested a task will be less satisfied than those participants who hear it attributed to Stephanie*, there was no main effect of Motive on the satisfaction ratings. However, there was an interaction effect. In the Near detour conditions, all ratings of the robot were relatively high, and the "Sorry, this will only take a minute" apology that everyone received might have been enough to mitigate dissatisfaction with the delay. However, in the Far condition, the Stephanie Motive resulted in significantly more satisfied participants than either of the other two Motives, partially supporting both H5 and H6. Future work is needed to determine if the reason that Stephanie was rated higher was because her office was the destination or because the explanation attributed the task to a specific person.

9 General Discussion

Together, our studies examined the effects of in-person interaction with a robot that performed multiple tasks with and without interleaving them. Inevitably, task interleaving meant that some people experienced real delays in task completion. Our first experiment varied task orders and durations while also teasing apart the differences in opinions between an observer watching the two tasks and the two task owners (i.e., people who requested the tasks). The second experiment varied task length for different participants while they all "owned" the guidance task. Across both studies, we found three patterns of effects for task interleaving: short delays do not have a major impact on perceptions of the robot, longer delays require mitigation techniques to maintain positive perceptions of the robot, and participants are more likely to endorse task interleaving in an abstract sense when they themselves will not experience the resulting delays. We also examined how the roles of the participants and the other task owners impact participant perceptions of the robot. We found that observers have more negative perceptions of the robot than task owners and that participants who own tasks are more likely to tolerate delays when they are for other known task owners.

When we specifically looked at perceptions of task interleaving for the robot, we found a relatively consistent set of results across our two studies. In both experiments, short delays did not have major impacts on perceptions of the robot. Specifically, the first experiment found that Requester 1 was not significantly dissatisfied with the detour path. This path added a delay to the fulfillment of their request, but the delay was relatively brief. In contrast, the sequential path introduced a very long delay to the completion of Requester 2's task and subsequently caused dissatisfaction. Notably, no incentives were provided for fast or efficient task completion. Generally, participants rated the robot taking at least one of the interleaved paths as more efficient, intelligent, knowledgeable, and humanlike than when it took the sequential path. The second experiment caused delays to a single participant and found that the shorter detour received more positive ratings than the farther detour, providing further support for the importance of delay lengths. Additionally, negative perceptions of the robot were reduced if a delay was justified. For Experiment 1, all of the participants could see that multiple people were making requests of the robot, with the result that all delays were justifiable. The degree of justification was then modulated by the delays. Experiment 2 found that participants who were not given a motive for their Far detours or only told the task was for "someone" rated the robot more negatively than those who were specifically told the task causing the Far detour was for "Stephanie," the person at their eventual destination. Finally, we found that participants were extremely likely to endorse interleaving tasks when it did not affect them directly. The results from the first experiment confirmed previous work by Carter et al. [4]: participants overwhelmingly recommended that the robot interleave tasks when asked in questions about imaginary scenarios.

9.1 The Roles of Task Owners and Observers and Implications for Online Research

It is important to those witnessing the robot's behaviors as well as those programming robots to know who owns various tasks. We found in our first experiment that the opinions of different task owners differ from each other, and they also differ from participants who are just observing the scene. Overall, participants rated the sequential Task Order more negatively on four characteristics relative to the various task interleaving conditions. Notably, there were also 10 items on which the observer rated the robot more negatively than one or both of the requesters did. These findings have implications for online human–robot interaction research. If a study is to be performed exclusively online, it is necessary to program tasks in such a way that participants stay engaged and potentially feel that the robot's behaviors impact them to avoid scoring patterns that may not reflect in-person experiences. In the second experiment, knowledge of who the other task owner was impacted participant perceptions as described above. Together, the results suggest that there is value in making a task queue with requester information available to everyone. Doing so could help clarify when tasks and subtasks are being efficiently interleaved as well as help justify long delays.

Additionally, our findings suggest that performing direct comparisons between in-person and online studies is important to ensure equivalencies in results. In our first experiment, the robot traveled comparable distances to those in the prior work by Walker et al. [27] and Carter et al. [4]. Detouring out of the way for a task was not found to impact ratings, similar to the two previous online studies. However, our second experiment integrated larger detours that required physical movement from participants. In that scenario, longer detours resulted in lower ratings of the robot. This difference may have arisen from the real-world impact of having to take a long detour as well as the participant's more active role in the task. It may also have been caused by the difference in tasks despite a consistent lack of incentivization for task completion. Thus, we recommend that researchers confirm the findings of online studies with in-person experiments when possible.

Notably, participants in both [4] and the current work consistently recommended interleaving tasks to make queues more efficient, even if that was not the behavior that they rated the highest when watching the robot. The contrast between the queue recommendations and the ratings suggests that the ratings may have not yet hit a ceiling: It is possible that we can continue to improve robot explanations and participant perceptions. Alternatively, it is possible that people are more likely to endorse task interleaving when it will not affect them directly, but they do not like it when it happens to them. Robot designers should experiment with varying explanations across a variety of contexts to see if they can maximize user experience. They should also consider for themselves whether a particular use case should prioritize efficiency versus human perceptions.

9.2 Limitations

There were some limitations to the current work. The experiments were limited in setting and scope. For Experiment 1, the items and information that were retrieved by the robot were not determined by the participants themselves. They neither wanted nor benefited from the tasks, and neither task was truly time-sensitive. They might have been more invested in the outcome in real life for tasks that they had elected. For Experiment 2, the participants were not in any hurry to get to their destination, but that was still a more urgent task than checking on a plant. Additionally, the robot did not report its findings to Stephanie upon arrival, potentially mitigating the impact of the task being performed for her. We expect that these scenarios are not fully analogous to the real world, but we were constrained by our desires to replicate previous work [4, 27] before further expanding upon it. The work still contributes a necessary step towards transferring online research to the field.

9.3 Future Work

For future work, one possible avenue is to explore further how being an observer (in person or remotely) compared to an active participant affects the positivity of the data as a whole. Within this research, it would also be possible to measure perceptions of the delays to see if participants with different roles have varying senses of the time that passes. Another possibility is to examine how people of various demographics perceive the robot and/or to perform longitudinal studies to determine how experience with robots affects participant perceptions. Anecdotally, some participants who revealed that they were very experienced with robots told us after the experiments that they preferred efficient paths, while novices were excited by the robot under all circumstances. Future research could specifically explore novelty and experience effects for various robots and participant groups. Additionally, the importance of different tasks (e.g., delivering something needed now versus something needed tomorrow) and incentives for participants to complete tasks quickly and/or efficiently could be manipulated to examine the impact on participant perceptions. Moreover, researchers could vary the practical impact of delays: a short delay might impact the person receiving an item differently if the item is an ice cream that will melt; it also might impact someone being guided differently if they are being made early versus late. Finally, the research could examine the impacts of task queuing and explanations in naturalistic settings like office buildings and hospitals to determine how robots should function in the field.

10 Conclusion

In conclusion, we have confirmed previous findings [4] that users of a robot generally recommend that it interleaves tasks in its queue for the sake of efficiency. However, it is important to minimize and justify any delays to a specific user's task to maintain positive impressions of the robot. Designers need to be careful to consider these competing needs when creating robot behaviors and determining how to queue tasks. Additionally, participants are more likely to endorse task interleaving in hypothetical scenarios when it does not affect them directly relative to when they experience the robot performing tasks in various orders and rate it. They also rate behaviors more negatively as observers (particularly online) relative to when they are involved in the process of requesting tasks. Therefore, we also recommend caution when generalizing results from studies where participants have varying levels in experience with and investment in the robot's behaviors.

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